MAJOR PROJECT · PHASE 2 PROGRESS PRESENTATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AgriAI 360

Beyond Crop Recommendation — Yield, Profit, Water, Risk, Disease, Fertilizer, Market Price, Notifications, AI Assistant & Real-Time Intelligence

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TEAM MEMBERS & PRESENTATION SPLIT

Alden Pais — NNM23CS020 (Yield & Profit Prediction)

Disha Shetty — NNM23CS067 (Water Requirement & Yield Risk)

Divya R — NNM23CS068 (Disease Detection & Fertilizer Advisor)

Harsheel S Gowda — NNM23CS082 (Market Price Intelligence & Notifications)

UNDER THE GUIDANCE OF

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NMAM Institute of Technology

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RECAP — PHASE 1

What Was Already Delivered

Phase 1 shipped the core, explainable crop-recommendation engine end to end: a trained Random Forest classifier over real soil + weather data, SHAP-based explanations, and a 4-model comparison lab — verified against a running system, not a mockup.

99.3%	Random Forest test accuracy on the 22-crop recommendation engine
SHAP	Real per-feature explanation for every "why this crop?" recommendation
4	Algorithms honestly compared (RF, Gradient Boosting, XGBoost, SVM)

THIS PRESENTATION COVERS WHAT'S BEEN BUILT SINCE

Ten new modules layered on top of that Phase 1 core — nine implemented and verified end-to-end (Yield, Profit, Water, Risk, Disease Detection, Fertilizer, Market Price, Notifications, and the AI Agriculture Assistant) and one partially delivered (Real-Time Agri Intelligence: Weather and IoT ingestion are real; Satellite/NDVI has no data provider connected) — all following the same rule: every number shown is real, trained, or from a published source — never fabricated.

PRESENTATION PLAN

8 Modules, One per Member Pair

Same format as the Phase 1 literature review — each member owns two modules end to end: the problem it solves, the method/data behind it, and the honest result.

MEMBER 1

Alden Pais

NNM23CS020

→ Yield Prediction (ML)

→ Profit Prediction

MEMBER 2

Disha Shetty

NNM23CS067

→ Water Requirement

→ Yield Risk (ML)

MEMBER 3

Divya R

NNM23CS068

→ Disease Detection / Crop Doctor (ML)

→ Fertilizer Advisor

MEMBER 4

Harsheel S Gowda

NNM23CS082

→ Market Price Intelligence

→ Personalized Notifications

+ 2 additional modules delivered as team-wide work (not assigned to an individual member): AI Agriculture Assistant · Real-Time Agri Intelligence

MODULE · PHASE 2

PRESENTED BY ALDEN PAIS

Yield Prediction

Estimates expected yield (tonnes/hectare) for a crop from its state, season and the trained regression model — a regional-historical estimate, not a per-farm forecast, and honestly labeled as such.

Dataset — 325,000 district-level India crop-production records, 1997–2020

Active model — XGBoost Regressor (also trained: Random Forest, Gradient Boosting)

Coverage — 11 of 22 crops — the rest honestly report "not available"

Shown — Right under the Top-5 crop recommendation on the detail page

MODEL PERFORMANCE

Algorithm	R^2	MAE
XGBoost (active)	0.919	1.03 t/ha
Random Forest	0.919	1.03 t/ha
Gradient Boosting	0.793	2.09 t/ha

HONEST LIMITATION

No access to an individual farm's soil, irrigation or this season's real weather — it answers "how has this crop performed in this state/season historically," not a per-farm prediction.

MODULE · PHASE 2

PRESENTED BY ALDEN PAIS

Profit Prediction

Not a trained model — a transparent calculation: real yield estimate × real historical market price, minus a cost the farmer supplies. Deliberately not modeled, because no honest per-farm cost-of-cultivation dataset exists.

Yield Estimate (t/ha) × 2016 Agmarknet Median Price = Revenue − Farmer-Supplied Cost = Estimated Profit

WHY COST ISN'T MODELED

No honest public per-farm cost-of-cultivation dataset was found. Inventing one would violate this project's own "never fabricate a number" rule — so cost stays farmer-supplied, by design.

COVERAGE

6 crops — the intersection of yield-supported crops and price-supported crops. Revenue uses real but non-inflation-adjusted 2016 mandi prices.

MODULE · PHASE 2

PRESENTED BY DISHA SHETTY

Water Requirement

Not a trained model — the real, published FAO formula, fetched and parsed directly from the primary FAO documents rather than recalled from memory.

ETo = p × (0.46 × Tmean + 8) → × FAO-56 Kc_mid − USDA effective rainfall = Net Irrigation Need

Coverage	17 of 22 crops
Output	mm/day and liters/day/acre

A BUG CAUGHT BEFORE SHIPPING

An earlier version derived p from solar day-length instead of using FAO's own table — it produced an unrealistic ETo ≈ 22.8 mm/day vs. the real ≈ 6.2 mm/day for a test case. Caught, documented, and replaced with FAO's actual source table.

MODULE · PHASE 2

PRESENTED BY DISHA SHETTY

Yield Risk (ML-Based)

A trained classifier predicting LOW / MEDIUM / HIGH — how a crop has historically ranked nationally for a given state/season. Augments, never replaces, the existing rule-based risk dashboard.

Model	Random Forest, 3 algorithms compared
Test accuracy	55.3% on 2016–2020 (years unseen in training) vs. 33.3% baseline
Sanity check	Punjab correctly predicted LOW risk for rice — India's best-known high-yield rice state

A TRAIN/TEST LEAKAGE BUG, CAUGHT

A first random-split attempt reported a flattering 64.5%. Proper cross-validation on the same data came back at only 36.9% — near-duplicate (state, crop, season) rows from different years were leaking across train/test. Fixed with a temporal split; real result: 55.3%.

MODULE · PHASE 3

PRESENTED BY DIVYA R

Disease Detection — "AI Crop Doctor"

Upload a leaf photo → real diagnosis. A frozen ImageNet-pretrained MobileNetV3-Small extracts a 576-dim feature vector; a trained classifier head identifies crop + disease/healthy condition.

Dataset 3,875 real leaf images (PlantVillage-Dataset)

Coverage 13 classes across 4 crops — apple, maize, grapes, orange

Active model Logistic Regression on MobileNetV3 features

Follow-up Real curated care guidance (pathogen, cultural + chemical control) per class

MODEL PERFORMANCE

Algorithm	Accuracy
Logistic Regression (active)	97.94%
Gradient Boosting	95.74%
Random Forest	95.48%

VERIFIED LIVE

Maize Common Rust: 99.8% confidence. Healthy apple leaf: 99.98% confidence — both through the real /api/predict/disease endpoint, not just offline metrics.

MODULE · PHASE 3

PRESENTED BY DIVYA R

Fertilizer Advisor

Rule-based decision support built on the same N/P/K/pH reference bands already used by the soil-health score — one source of truth, no drifting thresholds.

- Flags each nutrient DEFICIENT / ADEQUATE / EXCESS against agronomic reference ranges
- pH guidance for strongly acidic / alkaline soils (nutrient availability effects)
- Organic carbon (SOC) guidance — low / moderate / healthy bands
- Growth-stage-aware application timing: seedling → vegetative → flowering → fruiting → maturity
- Warns on excess nutrients and low soil moisture before application

DELIBERATELY NO DOSAGE NUMBER

Never outputs a kg/acre figure. Exact rates depend on a certified local soil-test report, crop variety, target yield and state agricultural-university guidance this system doesn't have — every response carries an explicit disclaimer.

MODULE · PHASE 3

Market Price Intelligence

PRESENTED BY HARSHEEL S GOWDA

Real Agmarknet (Government of India) mandi price records, queried through a swappable MarketPriceProvider interface — mirrors the same pattern already used for weather, so a live Agmarknet/eNAM feed can be swapped in later without touching routes or frontend.

Records 263,520 real daily mandi price rows, calendar year 2016

Coverage 6 crops across up to 23 states and 848 markets/crop

Capabilities Latest snapshot, price trend, cross-market comparison

SAMPLE DATA SCALE

Crop	Records	Markets
Rice	69,488	469
Maize	55,902	848
Pigeon peas	41,240	679

HONEST LABEL

Real government data, but a fixed historical year — clearly marked "not live" everywhere it's shown.

MODULE · PHASE 3

PRESENTED BY HARSHEEL S GOWDA

Personalized Notifications

Derives alerts from the same real data every other module already computes — soil health, weather, risk, latest disease diagnosis, market price trend. Never invents or randomizes content.

Soil Health → Weather → Risk → Disease Diagnosis → Market Trend → Deduplicated Farmer Alert

5 ALERT CATEGORIES

FERTILIZER (low soil score) · WEATHER (rain in forecast) · RISK (medium/high overall) · DISEASE (recent positive diagnosis) · MARKET (±5% price swing on the farmer's own top-recommended crop)

NO SPAM, BY DESIGN

A 12-hour dedup window skips re-creating an identical title+message, so repeated dashboard loads don't flood the farmer with duplicates.

MODULE · PHASE 3

AI Agriculture Assistant

A real, working chat grounded in the farmer's own data — not a general chatbot. Per the project's no-fabrication rule, it's built and explicitly labeled as a rule-based, keyword-classified Q&A engine, not an LLM.

No LLM key	No ANTHROPIC_API_KEY / OPENAI_API_KEY configured in .env.example
Grounded in	Real per-farm soil, weather, risk, disease, fertilizer, market price & live SHAP
Answers	Only recognized intents — says "I couldn't match that" rather than inventing
Persisted	Real chat_sessions / chat_messages database tables
Endpoint	POST /api/assistant/ask

SINGLE SWAP POINT

If a real LLM/RAG API is added later, answer_question() is the one call site to swap — the router and DB schema don't need to change.

VERIFIED LIVE, THIS SESSION

A live HTTP test against the running backend confirmed: a bare greeting gets a friendly reply without a farm selected; a real question without a farm is correctly asked to select one; and a real question with a farm returns an answer grounded in that farm's actual soil record. A guard bug (a substring match on "hi" that let unrelated messages like "why is my risk high" bypass the farm-selection check) was found and fixed in this same session.

MODULE · PARTIALLY IMPLEMENTED

Real-Time Agri Intelligence (Weather / Satellite / IoT)

Three sub-systems under one dashboard page — each honestly labeled for exactly what it is today. This is the one module, of the ten covered in this deck, that is not fully implemented.

WEATHER — real, dual-mode

Default is a deterministic mock provider (source: "mock"). A genuine live provider (Open-Meteo, free, no key) also exists and works — just not switched on by default.

SATELLITE / NDVI — not implemented, honestly so

GET /api/satellite/status always returns status: "PLANNED". No SATELLITE_API_KEY/SATELLITE_PROVIDER configured. Never fabricates an NDVI or vegetation-health reading.

IOT SENSORS — real ingestion, no hardware

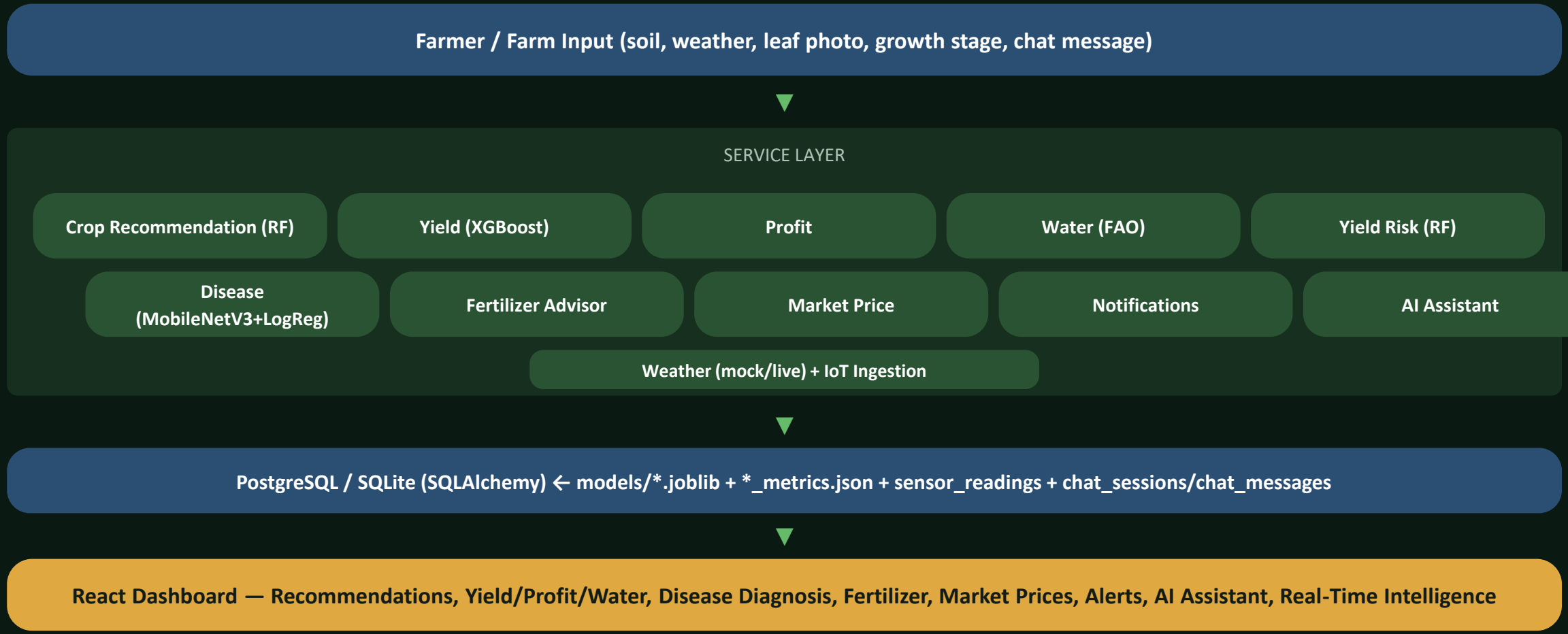
POST/GET /api/iot/readings are real, database-backed endpoints any physical gateway could use today. Demo readings (via /api/iot/demo/seed) are explicitly tagged source: "demo_simulation" — never mixed with real device data.

WHY THIS STAYS FUTURE SCOPE

Satellite/NDVI needs a real paid/registered data provider (Sentinel Hub, Earth Engine, or Planet) — not something this project can integrate without external access. Physical IoT sensors need actual hardware. Both are architecture-ready, not simulated as done.

SYSTEM ARCHITECTURE

Updated End-to-End Block Diagram



Not shown above: Satellite/NDVI — architecture-ready only, no data provider connected (GET /api/satellite/status always returns "PLANNED").

RESULTS & EVALUATION

Every Module's Real Number, Side by Side

Module	Method	Real Result	Crops Covered
Crop Recommendation	Random Forest	99.32% test accuracy	22
Yield Prediction	XGBoost Regressor	R^2 = 0.919, MAE = 1.03 t/ha	11
Profit Prediction	Revenue − Cost formula	Real 2016 mandi revenue	6
Water Requirement	FAO Blaney-Criddle	Real published formula + tables	17
Yield Risk	Random Forest classifier	55.3% test accuracy (33.3% baseline)	11
Disease Detection	MobileNetV3 + Logistic Regression	97.94% test accuracy	4 crops / 13 classes
Fertilizer Advisor	Rule-based reference bands	Same thresholds as soil health score	All
Market Price Intelligence	Real Agmarknet DB query	263,520 real mandi price records	6
Notifications	Rule-based aggregation	5 categories, 12h dedup window	All
AI Agriculture Assistant	Rule-based keyword classifier	Grounded in real per-farm data (not an LLM)	N/A
Real-Time Agri Intelligence	Weather: mock/live · Satellite: none · IoT: real ingestion	Weather & IoT real; Satellite always "PLANNED"	N/A

LIMITATIONS

Honest Constraints of the Current Build

YIELD / PROFIT / RISK

Regional-historical, not per-farm; profit's cost side is farmer-supplied, not modeled; risk covers only 11 of 22 crops.

WATER REQUIREMENT

Uses FAO's own "rough estimate" Blaney-Criddle method (not the more accurate Penman-Monteith); state-capital latitude, not farm GPS.

DISEASE DETECTION

Frozen-feature-extractor, not fine-tuned; only 4 crops; PlantVillage's lab-condition images, not India field-condition photos.

FERTILIZER / MARKET / NOTIFICATIONS

Fertilizer gives no dosage number by design; market prices are a fixed 2016 snapshot, not live; notifications only cover signals already computed elsewhere.

AI ASSISTANT

Rule-based by design, not a general-purpose LLM — recognizes a fixed set of intents and says so when a question doesn't match, rather than guessing.

REAL-TIME INTELLIGENCE

Weather defaults to mock data (a live provider exists but isn't default); Satellite/NDVI has no data provider connected; no physical IoT sensor is connected — the real ingestion pipeline runs on demo data only.

CURRENT IMPLEMENTATION

Overall Project Progress

9 / 10

Nine of the ten modules covered in this presentation — plus the full Phase 1 core — are implemented and verified end-to-end (backend test suite: 559/559 passing, plus a live end-to-end HTTP smoke test in this session). Real-Time Agri Intelligence is partially delivered: Weather and IoT ingestion are real; Satellite/NDVI has no data provider connected.

STILL NOT IMPLEMENTED

A real LLM/RAG-backed AI Assistant (today: a real, working rule-based version, by design), a voice interface, multilingual responses, a live Satellite/NDVI data provider, and physical IoT hardware (today: a real ingestion endpoint + demo data, no sensor connected).

NEXT STEPS

Optional polish only: switch WEATHER_PROVIDER to open-meteo for a live weather demo. Satellite and physical IoT integration remain intentionally out of scope — they require external paid providers / hardware this project doesn't have access to.

CONCLUSION

Summary & Path Forward

Phase 1 | Recommendation | Yield & Profit | Water & Risk | Disease Detection | Fertilizer & Market | Notifications | AI Assistant | Real-Time | Future Scope

AgriAI 360 has grown from a single explainable recommendation engine into a ten-module farm decision-support platform — nine implemented and verified end-to-end, one (Real-Time Agri Intelligence) partially delivered with Satellite/NDVI honestly left as future scope. Each module is trained, formula-verified, rule-based on real data, or built on real government/scientific data — and each one honest about what it doesn't yet know.

REFERENCES

Full reference list: 20 literature-review papers (Phase 1, slides 7-10) + project documentation (docs/dataset.md, docs/ml-methodology.md, docs/architecture.md, docs/future-scope.md) + each module's own train.py / service file + models/*_metrics.json + models/market_price_report.json + this session's code audit (backend test suite: 559/559 passing) and live end-to-end HTTP smoke test.